Safe System of Work

Securing Plant & Attachments to Trailers

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Overview

The following information provides guidance on the company's Standard Operation Procedures for works involving the securing of Plant, Machinery & Attachments to Vehicle Trailers prior to Transportation.

All employees have a responsibility for their safety and the safety of others by adhering to these codes of practice.

Hazards Identified

- Adverse weather
- Crushing
- Cuts and abrasions
- Vehicle movements
- Falls from height
- Manual handling
- Slips & Trips

Personal Protective Equipment (PPE) Requirements









This document provides guidance on the measures necessary for the safe carriage of tracked and wheeled engineering plant by vehicles constructed to comply fully with the Motor Vehicles (Construction and Use) Regulations 1986 and thereby permitted unrestricted use of the roads. It does not deal with the carriage of large machines etc. on special purpose vehicles whose use on the roads is restricted by current regulations. However, the general advice contained in this section will apply in many cases.

It is recommended that manufacturers of plant equipment either fit lashing points or provide a recommended lashing scheme for each of their vehicles.

Heavy engineering plant is normally transported on purpose-built vehicles which are specifically designed to provide easy loading and unloading facilities and are usually provided with adequate anchorage points for attaching the lashings. Lighter engineering plant may in some circumstances be carried on general-purpose vehicles. However, in these cases the method used to secure the load should provide equal security to that obtained by using purpose-built vehicles.

High loads may endanger bridges etc. over roads, so when these are carried it is essential that the driver know the exact height of the load, and the width of the load at that height. Also, loads with a high centre of gravity can seriously affect the vehicle's stability and such items of engineering plant should only be transported on vehicles with a low platform height.

A wheeled or tracked vehicle must be lashed down in position on the carrying vehicle, with the parking brake applied. The effectiveness of the parking brake on its own will be limited by the frictional resistance between the vehicle and the deck of the carrying vehicle, and even in normal driving conditions this will be inadequate and the vehicle will therefore require additional restraint. This additional restraint should take the form of a lashing system and some arrangement whereby the load is prevented from moving either forward or to the rear by an obstacle (or obstacles) securely fixed to the vehicle. These should butt against the wheels or tracks or some other part of the equipment carried.

All moveable assemblies such as jibs, brackets, booms and cabs etc. must be left in the position recommended for transportation by the manufacturer and must be secured to prevent movement relative to the main body of the machine.

Before the machine is moved onto the trailer all loose dirt that may otherwise come off and obstruct the highway or damage other vehicles must be removed. The ramp, the tyres of the machine and the bed of the trailer itself should all be free from oil, grease, ice etc. so that the machinery cannot slip. The transporter wheels should be chocked at the front and rear.

When the machine has been stowed and the engine stopped, the pressure in the hydraulic system should be relieved by moving all of the control levers through all their positions. This operation should be done at least twice. Controls should be set so as to prevent movement of ancillary items during transit.

Bags, tool kits, or other heavy objects should not be left loose in the operator's cab of the plant being carried.

The positioning of the engineering plant and any of its detached assemblies should be arranged so that the legal axle weight limits are not exceeded and the safe handling of the vehicle is not impaired. The clearance between the undersides of low loading vehicles and the road surface should be checked before moving off.

The machine should be positioned on the carrying vehicle's platform so that forward movement is prevented either by part of the main body of the vehicle, e.g. swan neck, step or headboard, or by an attached transverse member securely attached through the platform to the vehicle's chassis frame.

All items removed from the machine such as buckets, grabs, blades, shovels and lifting appliances should be lashed to the deck of the vehicle.

Wheeled and light tracked machines should be restrained so that the effect of bouncing caused by road shocks transmitted from the carrying vehicle and amplified by the machine's tyres or suspension units is minimised. Where possible the suspension unit of the machine should be locked and vertical movement limited by lashings or other means of restraint. Otherwise the machine's frame or chassis should be

supported on blocks.

Unless the machine is supported, the full contact area of its tyres, tracks, or drums should rest on the deck of the carrying vehicle. If the tracks extend outside the frame of the carrying vehicle then the machine's frame or chassis should be supported.

The machine should be restrained against forward, backward and sideways movement by chain or webbing lashings attached to anchorage points on the vehicle. All lashings should incorporate some form of tensioning device.

In deciding the number of anchorage points to be used when arranging a restraint system, the following factors should be considered:

- The need to position the machine to achieve the correct load distribution to meet the legal axle load requirements and to ensure that the vehicle's handling is not impaired.
- The extent to which other load restraint features is incorporated in the design of the vehicle.
- Whether the machine has wheels, tracks or rollers.
- The weight of the machine to be carried.
- A minimum of four separate anchorage points should be used.

The following guidelines apply to mobile engineering plant, that is vehicles fitted with hoists, working platforms, support legs and so on.

- High loads may endanger bridges and it is essential that the driver knows the height of the vehicle and has it displayed inside the vehicle cab.
- All moveable assemblies must be placed in the position and locked, where possible, as recommended for transportation by their manufacturer.

Restraining Devices

Apart from specialised fixing devices, the selection of materials for use in tie down schemes for engineering plant will be limited to chains, steel wire rope, webbing and their associated tensioning and coupling devices.

Where a transverse (side to side) beam is used as a baulk it should be securely fixed so that all loads imposed on it are transmitted to the carrying vehicle's chassis frame. Where individual wheels or drums are chocked with blocks or scotches these must be robust enough to resist crushing and be securely attached to the vehicle's platform where possible.

The lashings or securing devices should only be attached to those parts of the engineering plant, which are of sufficient strength to withstand the stresses likely to be imposed on them.

Where engineering plant comes equipped with dedicated lashing points for use when being transported, these points should be used and the vehicle secured as per manufacturer instructions. Care should be taken before lashing to lifting points as these may not be suitable for restraint purposes.

The loaded machine should be inspected after the vehicle has been driven for a short distance in order to check that no movement has taken place and that restraining devices are fully secure. Periodic inspections

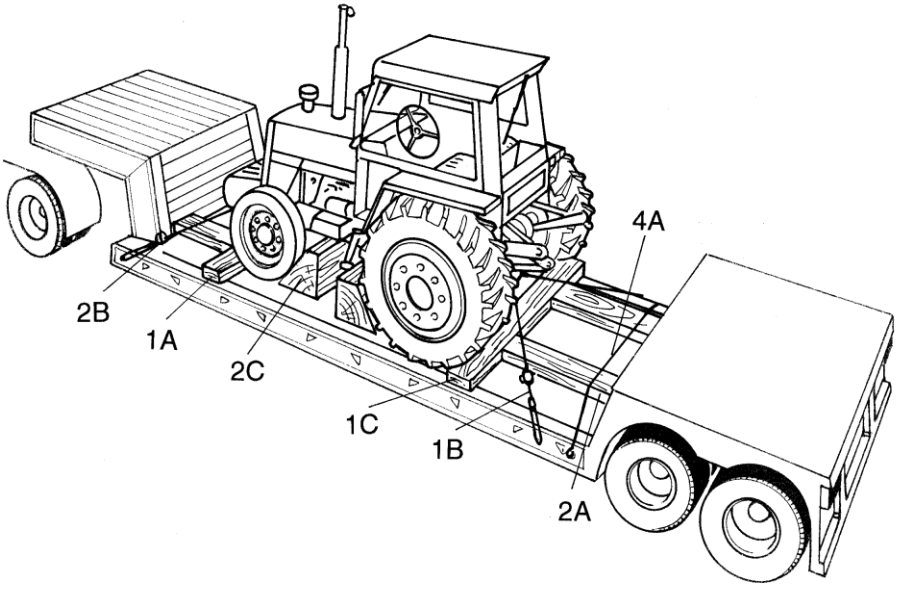
should be made during the course of the journey.

Suggested Tie Down Schemes

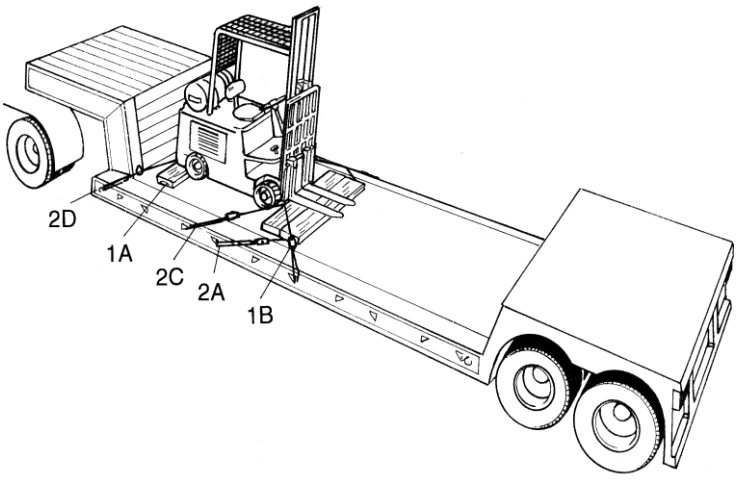
The tie down schemes described in the following diagrams and related schedules are typical systems which might be adopted for the securing of the various types of engineering plant. Variations to any of these schemes would be acceptable provided that all the basic safety precautions outlined earlier have been complied with.

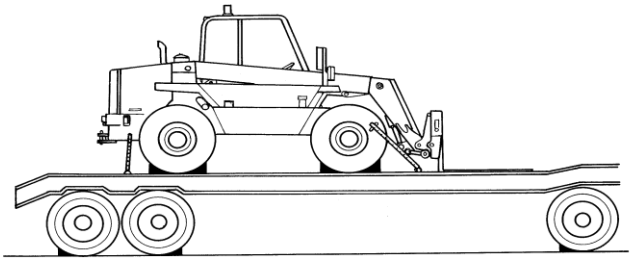
For simplicity, the figures show mainly chains used for lashings but in practice other lashing materials may be substituted.

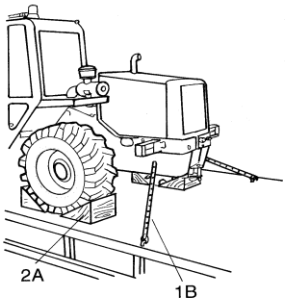
Tractors	**Control Measure / Precaution**
Possible hazard	
1. Forward movement of machine	**A.** Front wheels butted against bulkhead or against chocks against trailer bulkhead. **B.** Lashing chains from rear towing hook to anchorage points on trailer side members. **C.** Rear wheels butted against chocks.
2. Rearward movement of machine	**A.** Rear wheels butted against bulkhead or against chocks against trailer bulkhead. **B.** Lashing from front axle or towing hook to anchorage points on trailer side members. **C.** Front wheels butted against chocks.
3. Sideways movement of machine	**A.** Restraint provided by lashings used for forward and rearward restraint.
4. Movement of ancillaries	**A.** Lashings across bulking to anchorage points on trailer side members.

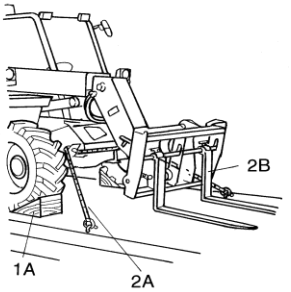


Fork Lift Trucks	Control Measure / Precaution
Possible hazard	
1. Forward movement of machine	**A.** Front and rear wheels of truck butted against chocks fixed to trailer bed.
	B. Truck lashed to trailer using manufacturers lashing points.
2. Rearward movement of machine	**A.** As **A.** and **B.** above.
	B. Forks lowered on to deck and hydraulic pressure relieved by operation of controls twice with engine switched off.
	C. Lashing from lashing point to anchorage points on trailer side members.
3. Sideways movement of machine	**A.** Restraint produced by lashings used for forward and rearward restraint.

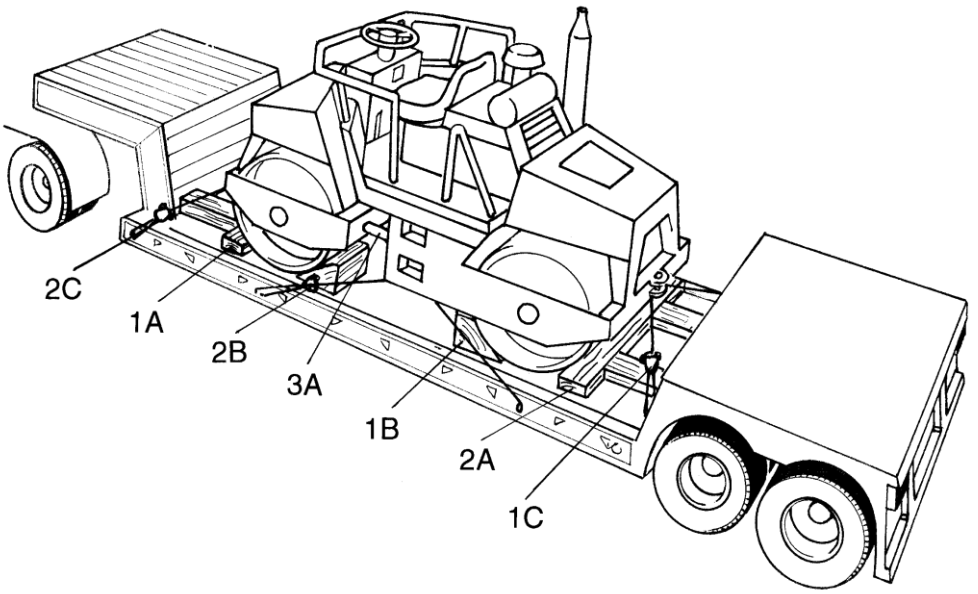




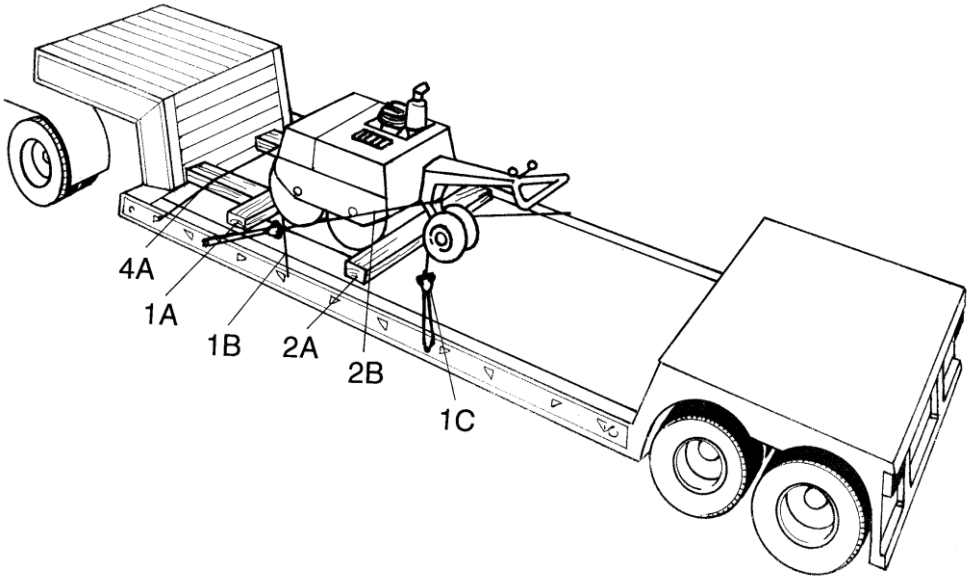




Ride-On Roller	Control Measure / Precaution
Possible hazard	
1. Forward movement of machine	**A.** Front roll butted against bulkhead or against chocks against trailer bulkhead.
	B. Lashing from frame and/or across member to anchorage points on trailer side members.
	C. Lashing from rear frame or towing hook to anchorage points on trailer side members.
2. Rearward movement of machine	**A.** Rear roll butted bulkhead or against chocks against trailer bulkhead.
	B. Lashing from frame and/or cross member to anchorage points on trailer side members.
	C. Lashing from front frame or towing hook to anchorage points on trailer side members.
3. Sideways movement of machine	**A.** On articulated equipment engaging the pivot lock mechanism will prevent articulation.
	B. Sideways movement will be prevented by the diagonal lashings used to prevent forward and rearward movement.

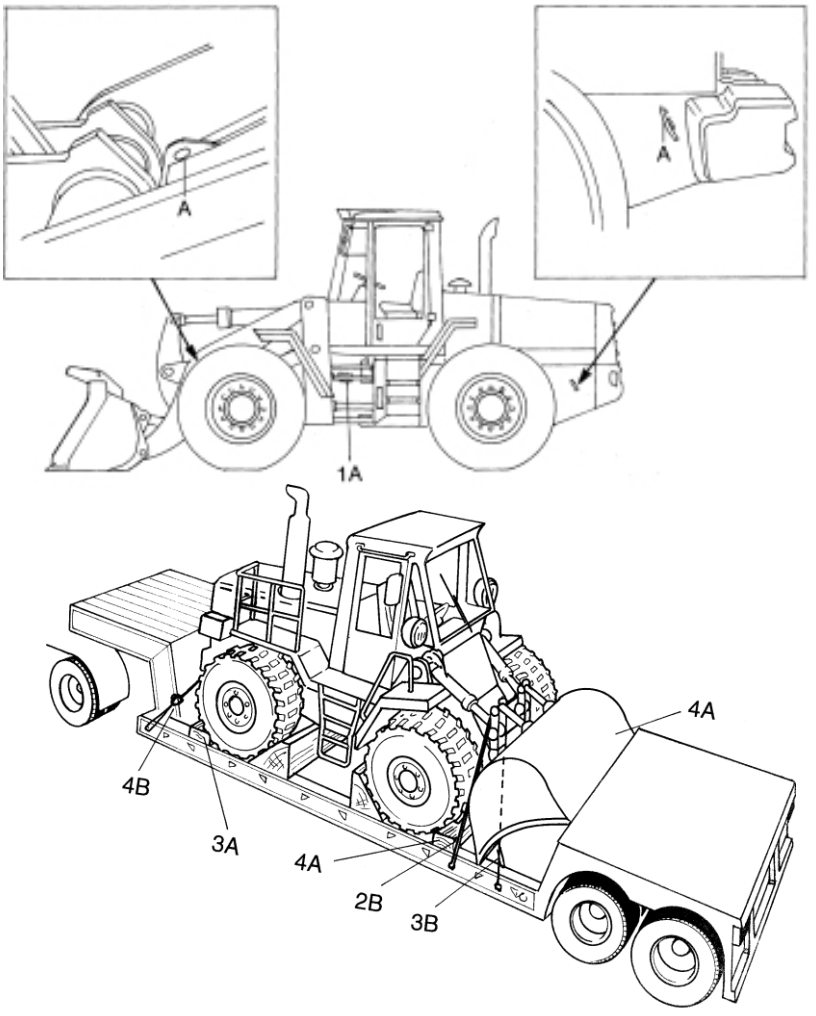


Towed Roller	Control Measure / Precaution
Possible hazard	
1. Forward movement of machine	**A.** Front roll butted against bulkhead or against chocks against trailer bulkhead. **B.** Lashing from and across the front of the machine body to anchorage points on trailer side members. **C.** Lashing around and across the rear of the rear wheel strut to anchorage points on trailer side members.
2. Rearward movement of machine	**A.** Rear roll butted against chocks. **B.** Lashing from and across the rear of the machine body to anchorage points on trailer side members.
3. Sideways movement of machine	**A.** Restraint provided by restraints to prevent forward and rearward movement.
4. Movement of ancillaries	**A.** Lashing across bulking.

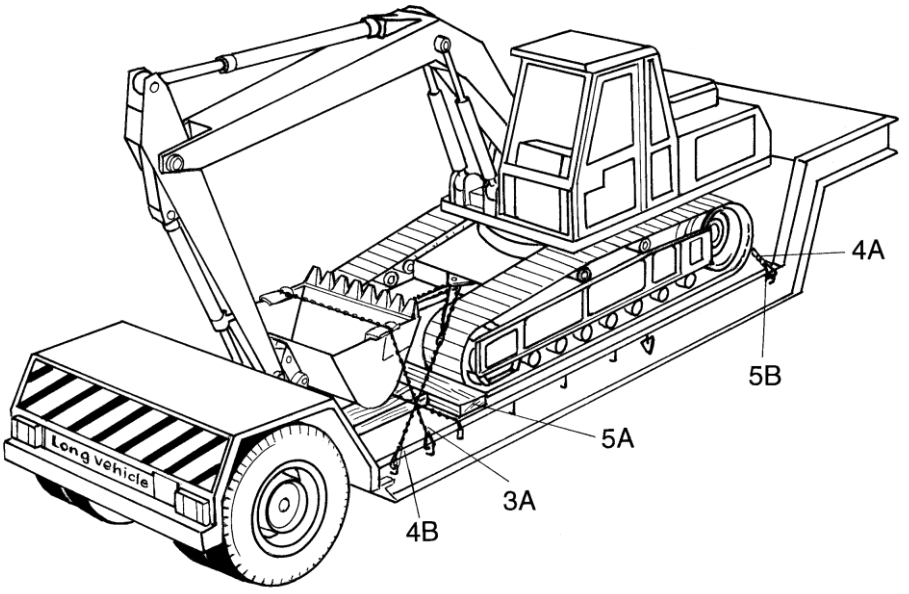


Wheeled Shovel

Possible hazard	Control Measures/Precautions
1. Articulation of front part of machine relative to rear part	**A.** Pivot locking bar bolted in position, in the case of pivot steer machines.
2. Movement of bucket assembly	**A.** Relieve hydraulic pressure in system by operating all controls twice, with the engine switched off. **B.** Lashings to secure bucket to the anchorage points on the trailer side members.
3. Forward movement of machine	**A.** Rear wheels butted against chocks against trailer bulkhead, or rear ballast weight butted against tractor bulkhead if strong enough. **B.** Lashing from front axle or towing hook to anchorage points on trailer side members.
4. Rearward movement of machine	**A.** Front wheels butted against chocks or bucket butted against rear bulkhead. **B.** Lashing from rear axle or towing hook to anchorage points on trailer side members.
5. Sideways movement of machine	**A.** Restraint provided by diagonal lashings used to prevent forward and rear movement.



Excavator (Tracked and Wheeled)	Control Measures/Precautions
Possible hazard	
1. Machine striking overhead obstruction	**A.** Stow equipment in position to give lowest overall height.
2. Movement of cab and superstructure relative to chassis of machine	**A.** Relieve hydraulic pressure by operating all controls twice, with engine switched off. **B.** Apply slew lock on slewing ring.
3. Movement of dipper arm away from stowed position	**A.** Lashing securing the bucket to anchorage points on the trailer side members.
4. Forward movement of machine	**A.** Tracks or wheels butted against the trailer bulkhead where weight distribution allows or butted against chocks. **B.** Lashing chains from excavator front towing point or chassis cross member to anchorage points on trailer side members.
5. Rearward movement of machine	**A.** Tracks or wheels butted against chocks. **B.** Lashing chains from excavator rear towing point or undercarriage lashing point to anchorage points on trailer side members.
6. Sideways movement of machine	**A.** Restraint provided by lashing chains used to prevent forward and rearward movement. Do not wedge heavy objects between the bucket and the machine chassis.
7. Movement of ancillaries	**A.** Lashings over baulking's.



Bulldozer	Control Measures/Precautions
Possible hazard	
1. Forward movement of machine	**A.** Bulldozer blade removed placed on front deck of trailer. **B.** Tracks butted against chocks against trailer bulkhead. **C.** Lashing from bulldozer blade U-frame trunnions to anchorage points on trailer side members.
2. Rearward movement of machine	**A.** Tracks butted against chocks against trailer bulkhead. **B.** Lashing from towing point to trailer side members.
3. Sideways movement of machine	**A.** Restraint provided by restraints to prevent forward and rearward movement. **B.** Lash from side members to trailer anchorage points.
4. Movement of ancillaries	**A.** Bulldozer blade stowed on front deck. Lashing across blade to trailer side members. **B.** Lashing across bulking.

